Perturbasyon

$$H\Psi = E\Psi \Rightarrow \quad H = \lambda^0 H_0 + \lambda H_1 + \dots \quad \Psi = \lambda^0\phi_n + \sum_{n\neq k}\lambda C_{nk}^1\phi_k + \sum_{n\neq k}\lambda^2 C_{nk}^2\phi_k + \dots$$

$$E = \lambda^0 E_n^0 + \lambda^1 E_n^1 + \lambda^2 E_n^2$$

$$\Rightarrow (H_0 + \lambda H_1)\Big(\phi_n + \sum_{n\neq k}\lambda C_{nk}^1\phi_k + \sum_{n\neq k}\lambda^2 C_{nk}^2\phi_k\Big) = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2)\Big(\phi_n + \lambda\sum_{n\neq k} C_{nk}^1\phi_k + \lambda^2\sum_{n\neq k} C_{nk}^2\phi_k\Big)$$

$$\Rightarrow H_0|\phi_n\rangle + \lambda\sum_{n\neq k} C_{nk}^1 H_0|\phi_k\rangle + \lambda^2\sum_{n\neq k} C_{nk}^2 H_0|\phi_k\rangle + \lambda H_1|\phi_n\rangle + \lambda^2\sum_{n\neq k} C_{nk}^1 H_1|\phi_k\rangle + \cancel{\lambda^2\sum_{n\neq k} C_{nk}^2 H_0|\phi_k\rangle} =$$

$$= E_n^0|\phi_n\rangle + \lambda\sum_{n\neq k} C_{nk}^1 E_n^0|\phi_k\rangle + \lambda^2\sum_{n\neq k} C_{nk}^2 E_n^0|\phi_k\rangle + \lambda E_n^1|\phi_n\rangle + \lambda^2\sum_{n\neq k} C_{nk}^1 E_n^1|\phi_k\rangle + \lambda^2 E_n^2|\phi_n\rangle$$

$\lambda^0 \Rightarrow H_0|\phi_n\rangle = E_n^0|\phi_n\rangle$

$\lambda^1 \Rightarrow \sum_{n\neq k} C_{nk}^1 H_0|\phi_k\rangle + H_1|\phi_n\rangle = \sum_{n\neq k} C_{nk}^1 E_n^0|\phi_k\rangle + E_n^1|\phi_n\rangle$

$\lambda^2 \Rightarrow \sum_{n\neq k} C_{nk}^2 H_0|\phi_k\rangle + \sum_{n\neq k} C_{nk}^1 H_1|\phi_k + \cancel{\sum_{n\neq k} C_{nk}^2 H_0|\phi_k\rangle} = \sum_{n\neq k} C_{nk}^2 E_n^0|\phi_k\rangle + \sum_{n\neq k} C_{nk}^1 E_n^1|\phi_k\rangle + E_n^2|\phi_n\rangle$

$\Rightarrow \langle\phi_n|$ ile genişlet;

$$\sum_{n\neq k} C_{nk}^1\langle\phi_n|H_0|\phi_k\rangle + \langle\phi_n|H_1|\phi_n\rangle = \sum_{n\neq k} C_{nk}^1 E_n^0 \cancel{\langle\phi_n|\phi_k\rangle}^{0} + E_n^1\langle\phi_n|\phi_n\rangle$$

$$\cancel{\langle\phi_n|E_k^0|\phi_k\rangle}$$

$\Rightarrow \langle\phi_n|H_1|\phi_n\rangle = E_n^1$

$\Rightarrow H_1|\phi_n\rangle = ? \quad \Rightarrow H_1|\phi_n\rangle = E_n^1|\phi_n\rangle \Rightarrow$

$$\Rightarrow \underbrace{\sum_{n\neq k} C_{nk}^1 H_0|\phi_k\rangle}_{\sum_{n\neq k} C_{nk}^1 E_k^0|\phi_k\rangle} + H_1|\phi_n\rangle = \sum_{n\neq k} C_{nk}^1 E_n^0|\phi_k\rangle + E_n^1|\phi_n\rangle$$

$\Rightarrow H_1|\phi_n\rangle = \sum_{n\neq k} C_{nk}^1(E_n^0 - E_k^0)|\phi_k\rangle + E_n^1|\phi_n\rangle \qquad \langle\phi_m|$ ile genişlet

$\Rightarrow \langle\phi_m|H_1|\phi_n\rangle = \sum_{n\neq k} C_{nk}^1(E_n^0 - E_k^0)\langle\phi_m|\phi_k\rangle + E_n^1\langle\phi_m|\phi_n\rangle \qquad |k\rangle \Rightarrow |n\rangle$, $\langle n| \Rightarrow \langle m|$

$\Rightarrow \langle\phi_m|H_1|\phi_n\rangle = \sum_{m\neq n} C_{mn}^1(E_m^0 - E_n^0)\langle\phi_m|\phi_n\rangle + E_n^1\langle\phi_m|\phi_n\rangle$

$$\Rightarrow \frac{\langle\phi_m|H_1|\phi_n\rangle}{E_m^0 - E_n^0} = C_{mn}^1$$

$$\sum_{n\neq k} C_{nk}^{2} H_0 |\phi_k\rangle + \sum_{n\neq k} C_{nk}^{1} H_1 |\phi_k\rangle + \sum_{n\neq k} C_{nk}^{2} H_0 |\phi_k\rangle = \sum_{n\neq k} C_{nk}^{2} E_n^{0} |\phi_k\rangle + \sum_{n\neq k} C_{nk}^{1} E_n^{1} |\phi_k\rangle + E_n^{2} |\phi_n\rangle$$

$\Rightarrow \langle \phi_k |$ ile genişlet

$$\sum_{n\neq k} C_{nk}^{2} \langle \phi_k | H_0 | \phi_k\rangle + \sum_{n\neq k} C_{nk}^{1} \langle \phi_k | H_1 | \phi_k\rangle = \sum_{n\neq k} C_{nk}^{2} E_n^{0} \langle \phi_k | \phi_k\rangle + \sum_{n\neq k} C_{nk}^{1} E_n^{1} \langle \phi_k | \phi_k\rangle + E_n^{2} \cancel{\langle \phi_k | \phi_n\rangle}$$

$$\sum_{n\neq k} C_{nk}^{2} E_k^{0} + \sum_{n\neq k} C_{nk}^{1} E_k^{1} = \sum_{n\neq k} C_{nk}^{2} E_n^{0} + \sum_{n\neq k} C_{nk}^{1} E_n^{1}$$

$$\sum_{n\neq k} C_{nk}^{2} H_0 |\phi_k\rangle + \sum_{n\neq k} C_{nk}^{1} H_1 |\phi_k\rangle + \cancel{\sum_{n\neq k} C_{nk}^{2} H_0 |\phi_k\rangle} = \sum_{n\neq k} C_{nk}^{2} E_n^{0} |\phi_k\rangle + \sum_{n\neq k} C_{nk}^{1} E_n^{1} |\phi_k\rangle + E_n^{2} |\phi_n\rangle$$

$\langle \phi_n | \Rightarrow$

$$\sum_{n\neq k} C_{nk}^{2} \underbrace{\langle \phi_n | H_0 | \phi_k\rangle}_{E_k^{0}|\phi_k\rangle} + \sum_{n\neq k} C_{nk}^{1} \langle \phi_n | H_1 | \phi_k\rangle = \sum_{n\neq k} C_{nk}^{2} E_n^{0} \underbrace{\langle \phi_n | \phi_k\rangle}_{0} + \sum_{n\neq k} C_{nk}^{1} E_n^{1} \underbrace{\langle \phi_n | \phi_k\rangle}_{0} + E_n^{2} \langle \phi_n | \phi_n\rangle$$

$$\frac{\langle \phi_m | H_1 | \phi_n\rangle}{E_m^{0} - E_n^{0}} = C_{mn}^{1}$$

$$\Rightarrow \sum_{n\neq k} C_{nk}^{1} \langle \phi_n | H_1 | \phi_k\rangle \Rightarrow \sum_{m\neq n} C_{mn}^{1} \langle \phi_m | H_1 | \phi_n\rangle = E_n^{2}$$

$$\Rightarrow \frac{|\langle \phi_m | H_1 | \phi_n\rangle|^2}{E_m^{0} - E_n^{0}} = E_n^{2} \qquad \langle \phi_n | H_1 | \phi_n\rangle = E_n^{1}$$

Problems

1. Consider the hydrogen atom, and assume that the proton, instead of being a point-source of the Coulomb field, is a uniformly charged sphere of radius R, so that the Coulomb potential is now modified to

$$V(r) = -\frac{3e^2}{2R^3}\left(R^2 - \frac{1}{3}r^2\right) \qquad r < R(\ll a_0)$$
$$= -\frac{e^2}{r} \qquad r > R$$

Calculate the energy shift for the $n = 1$, $l = 0$ state, and for the $n = 2$ states, caused by this modification, using the wave functions given in (12-25).

[4] Actually a simple barrier penetration calculation of the type carried out in Chapter 5 shows that the time scale is more like 10^{1000} lifetimes of the universe, for fairly reasonable fields!

$$\Rightarrow V(r) = -\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + e^2/r \qquad r < R \qquad \Rightarrow \quad -\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \qquad r < R$$

$$-e^2/r + e^2/r \qquad r > R \qquad\qquad 0 \qquad r > R$$

$$\Psi_{nlm} = N_n R_{nl}(r) Y(\theta,\phi) \Rightarrow \Psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

$$\text{Pert} \Rightarrow (H_0 + \lambda H_1) \Rightarrow E_n^{1} = \langle \phi_n | H_1 | \phi_n\rangle \Rightarrow \lambda \langle \phi_n | H_1 | \phi_n\rangle$$

$$\Rightarrow \lambda \int r^2 dr \int d\Omega \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot \left(-\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r}\right)$$

$$\lambda \int r^2 dr \int \underbrace{d\Omega}_{4\pi} \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot \left(-\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r}\right)$$

$$\Rightarrow \lambda \frac{4\cancel{\pi}}{\cancel{\pi} a_0^3} \int r^2 e^{-2r/a_0} \left(-\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r}\right) dr \Rightarrow$$

$$\Rightarrow \frac{\lambda 4}{a_0^3}\left(-\frac{3e^2}{2R}\right) \underbrace{\int_0^R r^2 e^{-2r/a_0} dr}_{(i)} + \frac{\lambda 4}{a_0^3}\left(\frac{e^2}{2R^3}\right) \underbrace{\int_0^R r^4 e^{-2r/a_0} dr}_{(ii)} + \underbrace{\frac{\lambda 4 e^2}{a_0^3}\int_0^R r e^{-2r/a_0} dr}_{(iii)}$$

$$\Rightarrow \Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt \Rightarrow 2r/a_0 = t \Rightarrow r = \frac{a_0 t}{2} \Rightarrow dr = \frac{a_0}{2} dt$$

(i) $$\Rightarrow -\frac{6e^2}{\cancel{a_0^3} R} \int \frac{\cancel{a_0}}{2} dt \cdot e^{-t} \frac{\cancel{a_0^2} t^2}{4} \Rightarrow -\frac{6e^2}{R8} \int \underbrace{e^{-t} t^2 dt}_{\Gamma(3)=2!} \Rightarrow -\frac{12e^2}{8R} = -\frac{3e^2}{2R}$$ //

(ii) $$\frac{\cancel{4}\lambda}{\cancel{a_0^3}}\left(\frac{e^2}{R^3}\right) \int \frac{a_0^{\cancel{4}1} t^4}{16} e^{-t} \frac{a_0}{\cancel{2}} dt \Rightarrow \frac{e^2 a_0^2}{16 R^3} \int \underbrace{t^4 e^{-t} dt}_{\Gamma(5)=4!} \Rightarrow \frac{24}{16} \frac{e^2 a_0^5}{R^3} = \frac{3}{2} \frac{e^2 a_0^2}{R^3}$$

(iii) $$\frac{\lambda 4 e^2}{a_0^3} \int r e^{-2r/a_0} dr \Rightarrow \frac{\cancel{4}\lambda e^2}{\cancel{a_0^3}} \int \frac{\cancel{a_0} t}{\cancel{2}} e^{-t} \frac{\cancel{a_0}}{\cancel{2}} dt \Rightarrow \frac{\lambda e^2}{a_0} \int \underbrace{t e^{-t} dt}_{\Gamma(2)=1} = \frac{\lambda e^2}{a_0}$$

(i) + (ii) + (iii) $$\lambda\left(-\frac{3e^2}{2R} + \frac{3}{2}\frac{e^2 a_0^2}{R^3} + \frac{e^2}{a_0}\right)$$

Quantum Physics - S. Gasiorowicz

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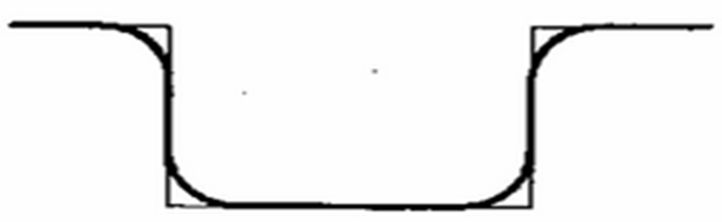
2. Calculate the energy shift in the ground state of the one-dimensional harmonic oscillator, when the perturbation

$$V = \lambda x^4 \qquad V = \lambda x^2$$

is added to

$$H = \frac{p^2}{2m} + \tfrac{1}{2}m\omega^2 x^2$$

3. Consider a square well in one dimension. If the edges of the well are rounded off as shown in the figure, what is the change in the ground state energy? Choose your rounding-off parametrization such that $\int_{-\infty}^{\infty} V(x)\,dx$ remains unchanged.



$$\Rightarrow E_n^1 = \lambda\langle\psi|H_1|\psi\rangle \qquad \frac{|\langle\phi_m|H_1|\phi_n\rangle|^2}{E_m^0 - E_n^0} = E_n^2 \qquad \langle\phi_n|H_1|\phi_n\rangle = E_n^1$$

$$\Rightarrow A = \sqrt{\frac{m\omega}{2\hbar}}x - i\frac{p}{\sqrt{2m\omega\hbar}} \qquad A^+ = \sqrt{\frac{m\omega}{2\hbar}}x + i\frac{p}{\sqrt{2m\omega\hbar}} \Rightarrow A + A^+ = 2\sqrt{\frac{m\omega}{2\hbar}}x$$

$$\Rightarrow x = (A + A^+)\frac{1}{2}\sqrt{\frac{2\hbar}{m\omega}} \Rightarrow (A + A^+)\sqrt{\frac{\hbar}{2m\omega}}$$

$X = (A+A^+)\frac{1}{2}\sqrt{\frac{2\hbar}{m\omega}} \Rightarrow (A+A^+)\sqrt{\frac{\hbar}{2m\omega}}$ $\qquad [A, A^+] = \mathbb{1} \Rightarrow AA^+ - A^+A = \mathbb{1}$

$AA^+ = \mathbb{1} + A^+A$

$$X^2 = \frac{\hbar}{2m\omega}(A+A^+)(A+A^+)$$

$$= \frac{\hbar}{2m\omega}(A^2 + AA^+ + A^+A + A^{+2})$$ λx^2 pert

$E_n^1 = \lambda\langle\psi|H_1|\psi\rangle \qquad E_n^2 = \lambda\frac{|\langle\psi_m|H_1|\psi_n\rangle|^2}{E_m^0 - E_n^0}$

$A|n\rangle = \sqrt{n}|n-1\rangle \qquad A^+|n\rangle = \sqrt{n+1}|n+1\rangle$

$\langle x^2\rangle \Rightarrow \langle n|\lambda x^2|n\rangle \Rightarrow \lambda\langle n|\frac{\hbar}{2m\omega}(A^2+AA^++A^+A+A^{+2})|n\rangle$

$\Rightarrow \lambda\frac{\hbar}{2m\omega}\{\langle n|A^2|n\rangle + \langle n|AA^+|n\rangle + \langle n|A^+A|n\rangle + \langle n|A^{+2}|n\rangle\}$

$\Rightarrow \lambda\frac{\hbar}{2m\omega}\{\langle n|A\sqrt{n}|n-1\rangle + \langle n|A\sqrt{n+1}|n+1\rangle + \langle n|A^+\sqrt{n}|n-1\rangle + \langle n|A^+\sqrt{n+1}|n+1\rangle$

$\cancel{\langle n|\sqrt{n}\sqrt{n-1}|n-2\rangle} \qquad \langle n|n+1|n\rangle \qquad \langle n|n|n\rangle \qquad \langle n|\sqrt{n+1}\sqrt{n+2}|n+2\rangle$

$n+1\langle n|n\rangle \qquad n\langle n|n\rangle \qquad \sqrt{n+1}\sqrt{n+2}\,\cancel{\langle n|n+2\rangle}^{\,0}$

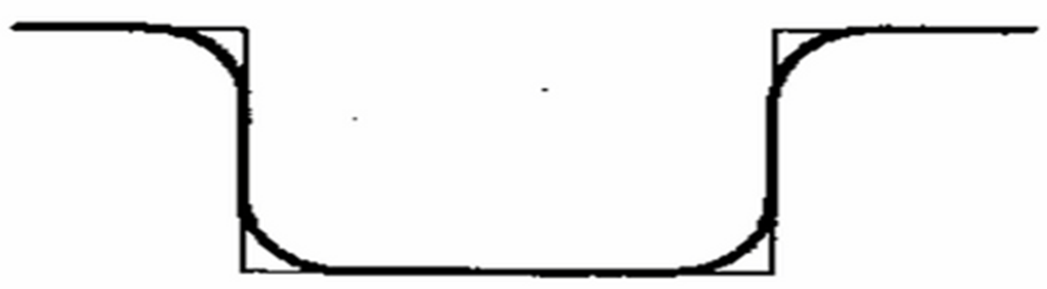
$n+1 \qquad n$

$$\underbrace{\qquad\qquad\qquad\qquad}_{2n+1}$$

$\Rightarrow \lambda\frac{\hbar}{2m\omega}(2n+1) \Rightarrow \frac{\lambda\hbar}{\cancel{2}m\omega}(\cancel{2})(n+\frac{1}{2}) \Rightarrow \frac{\lambda\hbar}{m\omega}(n+\frac{1}{2}) \Rightarrow \hbar\omega_0(n+\frac{1}{2})$

harmonik $E_n = \hbar\omega(n+\frac{1}{2})$

changed.



4. The bottom of an infinite well is changed to have the shape

$$V(x) = \epsilon \sin\frac{\pi x}{b} \qquad 0 \le x \le b$$

Calculate the energy shifts for all the excited states to first order in ϵ. Note that the well originally had $V(x) = 0$ for $0 \le x \le b$, with $V = \infty$ elsewhere.

$\Rightarrow \quad u(x) = u_n^0 = \sqrt{\frac{2}{b}}\sin\frac{n\pi x}{b} \qquad E_n = E_n^0 + \lambda E_n^1 + \dots$

$\Rightarrow E_n^1 = \langle n|H_1|n\rangle \Rightarrow E_n^1 = \langle u_n^0|H_1|u_n^0\rangle = \int_0^b dx\sqrt{\frac{2}{b}}\sin\frac{n\pi x}{b}\cdot\epsilon\sin\frac{\pi x}{b}\sqrt{\frac{2}{b}}\sin\frac{n\pi x}{b}$

$\frac{2\epsilon}{b}\int dx\,\sin^2\frac{n\pi x}{b}\cdot\sin\frac{\pi x}{b} \Rightarrow \frac{\pi x}{b} = y \Rightarrow dx = \frac{b\,dy}{\pi} \Rightarrow \frac{2\epsilon}{\cancel{b}}\frac{\cancel{b}}{\pi}\int_0^\pi dy\,\sin^2 ny\,\sin y$

Soru $H = \begin{pmatrix}1&0&0\\0&3&0\\0&0&-2\end{pmatrix} + \begin{pmatrix}0&c&0\\c&0&0\\0&0&c\end{pmatrix}$

a) H' nın özdeğerleri

b) pert. teorisi ile 2. mert. kadar düzeltmeler

a) $H\psi = E\psi \Rightarrow H = \begin{pmatrix}1&c&0\\c&3&0\\0&0&c-2\end{pmatrix} \Rightarrow \det(\mathbb{1}-\lambda) \Rightarrow H \Rightarrow \begin{vmatrix}1-\lambda & c & 0\\ c & 3-\lambda & 0\\ 0&0&c-2-\lambda\end{vmatrix}$

$\begin{vmatrix}1-\lambda & c & 0\\ c & 3-\lambda & 0\\ 0&0&c-2-\lambda\end{vmatrix} \Rightarrow \det = (c-2-\lambda)(-1)^{3+3}\begin{vmatrix}1-\lambda & c\\ c & 3-\lambda\end{vmatrix} = 0 \Rightarrow$

$\Rightarrow \underline{(c-2-\lambda)}.\{(3-\lambda).(1-\lambda) - c^2\} = 0 \quad \lambda_1\ \lambda_2\ \lambda_3$

$\lambda_1 = c-2 \qquad \lambda_{2,3} = ?$

$\Rightarrow \lambda_{2,3} \Rightarrow \lambda^2 - 4\lambda + 3 - c^2 \qquad \Delta = b^2 - 4ac \qquad \dfrac{-b \mp \sqrt{\Delta}}{2a}$

$\Rightarrow \dfrac{+4 \mp \sqrt{16-12+4c^2}}{2} \Rightarrow \dfrac{+4 \mp 2\sqrt{c^2+1}}{2} \Rightarrow$

$\lambda_{2,3} = 2 \mp \sqrt{c^2+1} \qquad \lambda_1 = c-2$

b) $E_n^1 = \lambda\langle\psi|H_1|\psi\rangle \qquad E_n^2 = \lambda\dfrac{|\langle\psi_m|H_1|\psi_n\rangle|^2}{E_m^0 - E_n^0}$

$H_1|\phi_n\rangle = E_n|\phi_n\rangle \Rightarrow \langle\phi_m|H_1|\phi_n\rangle = \underline{\langle\phi_m|E_n|\phi_n\rangle}$ $\quad i\ j \quad E_n\delta_{mn}$

$H = \underbrace{\begin{pmatrix}1&0&0\\0&3&0\\0&0&2\end{pmatrix}}_{H_0} + \underbrace{\begin{pmatrix}0&c&0\\c&0&0\\0&0&c\end{pmatrix}}_{H_1}$

$\begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & & \\ \hline m \rightarrow 1 & E_1 & 0 & 0 & 0 & \cdots & \\ m \rightarrow 2 & 0 & E_2 & 0 & 0 & \cdots & \\ 3 & 0 & 0 & E_3 & 0 & 0 & \cdots \\ 4 & \vdots & \vdots & & E_4 & \cdots & \end{array}$

$(H_0 + \lambda H_1) \Rightarrow E_n^0 + \lambda E_n^1 + \ldots$

$E_1^0 + E_1^1 + \ldots \qquad E_n^0 = \begin{pmatrix}1&0&0\\0&3&0\\0&0&2\end{pmatrix} \Rightarrow \begin{pmatrix}E_{11}^0 & E_{12}^0 & E_{13}^0\\ E_{21}^0 & E_{22}^0 & E_{23}^0\\ E_{31}^0 & E_{32}^0 & E_{33}^0\end{pmatrix} \Rightarrow E_{11}^0 = 1, \quad E_{22}^0 = 3, \quad E_{33}^0 = 2$

$E_n^1 = \lambda\langle\psi|H_1|\psi\rangle \Rightarrow E_n^1 = \begin{pmatrix}0&c&0\\c&0&0\\0&0&c\end{pmatrix} \Rightarrow \begin{pmatrix}E_{11}^1 & E_{12}^1 & E_{13}^1\\ E_{21}^1 & E_{22}^1 & E_{23}^1\\ E_{31}^1 & E_{32}^1 & E_{33}^1\end{pmatrix} \quad E_{11}^1 = 0, \quad E_{22}^1 = 0, \quad E_{33}^1 = c$

$E_n^2 = \lambda\dfrac{|\langle\psi_m|H_1|\psi_n\rangle|^2}{E_m^0 - E_n^0} \Rightarrow$

$$E_n^2 = \sum \frac{|\langle \psi_m | H_1 | \psi_n \rangle|^2}{E_m^0 - E_n^0} \Rightarrow E_{ni}^2 = \sum \frac{|\langle \psi_{mi} | H_1 | \psi_{ni} \rangle|^2}{E_m^0 - E_n^0}$$

$$i = n \Rightarrow \sum \frac{|\langle \psi_{mi} | H_1 | \psi_{ii} \rangle|^2}{E_m^0 - E_n^0}$$

$$\sum_{m \neq i} \frac{(H_1)_{mi} (H_1)_{im}}{E_m^0 - E_i^0} \qquad \begin{pmatrix} E_{11}^2 & E_{12}^2 & E_{13}^2 \\ E_{21}^2 & E_{22}^2 & E_{23}^2 \\ E_{31}^2 & E_{32}^2 & E_{33}^2 \end{pmatrix} \qquad \begin{matrix} E_{11}^2 \\ E_{22}^2 \\ E_{33}^2 \end{matrix}$$

$$\Rightarrow E_{11}^2 = \sum_{\substack{m \neq i \\ i=2}}^{3} \frac{(H_1)_{12} (H_1)_{21}}{E_1^0 - E_2^0} + \sum_{\substack{m \neq i \\ i=3}}^{3} \frac{(H_1)_{13} (H_1)_{31}}{E_1^0 - E_3^0} \qquad H = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}}_{H_0} + \underbrace{\begin{pmatrix} 0 & c & 0 \\ c & 0 & 0 \\ 0 & 0 & c \end{pmatrix}}_{H_1}$$

$$\frac{c^2}{1-3} + \frac{0.0}{1-2} = -\frac{c^2}{2} = E_{11}^2$$

$E_{11}^0 = 1$ $\quad E_{33}^0 = 2$

$E_{22}^0 = 3$

$$E_{22}^2 = \sum_{\substack{m \neq i \\ i=1}} \frac{(H_1)_{21} (H_1)_{12}}{E_2^0 - E_1^0} + \sum_{\substack{m \neq i \\ i=3}} \frac{(H_1)_{23} (H_1)_{32}}{E_2^0 - E_3^0} = \frac{c.c}{3-1} + \frac{0.0}{3-2} = \frac{c^2}{2}$$

$m=2$

$E_{11}^1 = 0$

$E_{22}^1 = 0$

$E_{33}^1 = c$

$$E_{33}^2 \Rightarrow \sum_{\substack{m \neq i \\ i=1}} \frac{(H_1)_{31} (H_1)_{13}}{E_3^0 - E_1^0} + \sum_{\substack{m \neq i \\ i=2}} \frac{(H_1)_{32} (H_1)_{23}}{E_3^0 - E_2^0} = \frac{0.0}{2-1} + 0 = 0$$

$m=3$

$$\underline{E_{22}^1} = -\frac{c^2}{2} \qquad \underline{E_{22}^2} = \frac{c^2}{2} \qquad \underline{E_3^{22}} = 0$$